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Problem 1.3

Using Dirac delta functions in the appropriate coordinates, express the following as 3D charge densities $\rho(\vec{x})$

(a) spherical coordinates, charge Q uniformly distributed over a spherical shell of radius R.

$$\begin{aligned}
\rho(\vec{x}) &= \rho(r,\theta,\phi) = \rho(r) = A\,\delta(r-R) \\
d^3x &= r^2 \sin\theta\, dr\, d\theta\, d\phi \quad \text{(volume element in spherical coords)} \\
\int d^3x'\,\rho(\vec{x}') &= \left[\int_0^r r'^2\,\rho(r')\,dr'\right]\left[\int_0^\pi \sin\theta'\,d\theta'\right]\left[\int_0^{2\pi} d\phi'\right], \quad r > R \\
Q &= \left[A\int_0^r r'^2\,\delta(r'-R)\,dr'\right][2]\,[2\pi] = 4\pi\,R^2\,A \\
\rightarrow A &= \frac{Q}{4\pi\,R^2}
\end{aligned}$$

so we have

$$\boxed{\rho(r) = \sigma_a\,\delta(r-R), \quad \sigma_a = \frac{Q}{4\pi\,R^2}}$$

Note that the radial delta function has dimension 1/Length since

$$\int_0^r \delta(x-R)\,dx = 1, \quad r > R$$

so the charge density has the correct dimensions (charge per unit volume).

(b) cylindrical coordinates, charge per unit length λ uniformly distributed over a cylindrical surface of radius b.

$$\begin{aligned}
\rho(\vec{x}) &= \rho(s,\phi,z) = \rho(s) = A\,\delta(s-b) \\
d^3x &= s\,ds\,d\phi\,dz \quad \text{(volume element in cylindrical coords)} \\
\int d^3x'\,\rho(\vec{x}') &= \left[\int_0^s s'\,\rho(s')\,ds'\right]\left[\int_0^{2\pi} d\phi'\right]\left[\int_z^{z+L} dz'\right], \quad s > b \\
\lambda\,L &= \left[A\int_0^s s'\,\delta(s-b)\,ds'\right][2\pi]\,[L] = 2\pi\,A\,L\,b \\
\rightarrow A &= \frac{\lambda}{2\pi\,b}
\end{aligned}$$

where we have used the symbol s for the radius in cylindrical coordinates to avoid confusion with the charge density (there are only so many letters in the Greek alphabet) and took a section of length L to compute the total charge

$$\boxed{\rho(s) = \sigma_b\,\delta(s-b), \quad \sigma_b = \frac{\lambda}{2\pi\,b}}$$

(c) cylindrical coordinates, charge Q spread uniformly over a flat circular disc (negligible thickness) with radius R. [at position z_0 along the z-axis]

$$\rho(s,\phi,z) = \rho(s,z) = A\,[\Theta(s) - \Theta(s-R)]\;\delta(z-z_0)$$

$$\text{step function } \Theta(x) = \begin{cases} 0, & x<0 \\ 1, & x>1 \end{cases}$$

$$\Theta(s) - \Theta(s-R) = \begin{cases} 1, & 0<s<R \\ 0, & s>R \end{cases}$$

$$\int d^3x'\,\rho(\vec{x}') = \left[A\int_0^R s'\,ds'\right]\left[\int_0^{2\pi} d\phi'\right]\left[\int_{z_1}^{z_2} dz'\,\delta(z-z')\right]$$

$$Q = \left[A\,\frac{R^2}{2}\right][2\pi]\,[1]\,,\quad z_1<z_0<z_2$$

$$\rightarrow A = \frac{Q}{\pi R^2}$$

where the integration runs over a volume that includes the entire disk,

$$\rho(s,z) = \sigma_c\,[\Theta(s) - \Theta(s-R)]\;\delta(z-z_0),\quad \sigma_c = \frac{Q}{\pi R^2}$$

which we can also simplify to

$$\boxed{\rho(s,z) = \sigma_c\,\Theta(R-s)\;\delta(z-z_0)}$$

since the radial coordinate s is by definition non-negative.

(d) same as (c) but in spherical coordinates; let's start by putting the disk in the x-y plane ($z_0=0$) for simplicity

$$\rho(r,\theta,\phi) = \rho(r,\theta) = A\,\Theta(R-r)\;\delta\left(\theta-\frac{\pi}{2}\right)$$

$$\int d^3x'\,\rho(\vec{x}') = \left[\int_0^r r'^2\,\rho(r')\,dr'\right]\left[\int_0^\pi \sin\theta'\,d\theta'\right]\left[\int_0^{2\pi} d\phi'\right],\quad r>R$$

$$= A\left[\int_0^R r'^2\,dr'\right]\left[\sin\left(\frac{\pi}{2}\right)\right][2\pi]$$

$$Q = A\left[\frac{R^3}{3}\right][1]\,2\pi$$

$$\rightarrow A = \frac{3}{2\pi}\,\frac{Q}{R^3}$$

leading to

$$\boxed{\rho(r,\theta) = \rho_d\,\Theta(R-r)\;\delta\left(\theta-\frac{\pi}{2}\right),\quad \rho_d = \frac{3}{2\pi}\,\frac{Q}{R^3}}$$

For the more general case with $z_0 \neq 0$, the disk is described by a constraint that relates the polar angle θ and the radius r

$$r\,\cos\theta = z_0$$

and the ranges for those coordinates are

$$\theta \in [0,\theta_0]\,,\quad \tan\theta_0 = \frac{R}{z_0}$$

$$r \in [z_0, r_0]\,,\quad r_0 = \sqrt{R^2+z_0^2} = R\sqrt{1+\left(\frac{z_0}{R}\right)^2}$$

The charge density can then be written as

$$\rho(\vec{x}) = \rho(r,\theta) = A\,\Theta\left(\theta_0 - \theta\right)\,\delta\left(r - \frac{z_0}{\cos\theta}\right)$$

Assuming $z_0 > 0$,

$$\begin{aligned}
\int d^3x'\,\rho(\vec{x}') &= \left[\int_0^r dr' \int_0^\pi d\theta'\, r'^2 \sin\theta'\,\rho(r',\theta')\right]\left[\int_0^{2\pi} d\phi'\right], \quad r > R \\
&= \left[A\int_0^r dr' \int_0^{\theta_0} d\theta'\, r'^2 \sin\theta'\,\delta\left(r' - \frac{z_0}{\cos\theta'}\right)\right][2\pi] \\
&= \left[A\int_0^{\theta_0} d\theta'\,\sin\theta' \int_0^r dr'\, r'^2\,\delta\left(r' - \frac{z_0}{\cos\theta'}\right)\right]\,2\pi \\
\frac{Q}{2\pi\,A} &= \int_0^{\theta_0} d\theta'\,\sin\theta'\left(\frac{z_0}{\cos\theta'}\right)^2 \\
&= z_0^2\int_{\cos\theta_0}^1 u^{-2}\,du = z_0^2\,\left(-\frac{1}{u}\right)\Bigg|_{\cos\theta_0}^1 \\
&= z_0^2\left(\frac{R}{z_0} - 1\right) = z_0(R - z_0) \\
\rightarrow A &= \frac{1}{2\pi}\frac{Q}{z_0(R - z_0)}
\end{aligned}$$

which gives

$$\boxed{\rho(r,\theta) = \sigma_d\,\Theta\left(\theta_0 - \theta\right)\,\delta\left(r - \frac{z_0}{\cos\theta}\right), \quad \sigma_d = \frac{1}{2\pi}\frac{Q}{z_0\,R\,\left(1 - \frac{z_0}{R}\right)}}$$

The dimensions are fine (Q/L^2 for σ and $1/L$ for the δ-function in r) but we can't take the limit $z_0 \to 0$ of this expression as a crosscheck. We could have also chosen a different form of the δ-function to confine the charge density to the surface of the disk

$$\begin{aligned}
\rho(r,\theta) &= A\,\left[\Theta(r - z_0) - \Theta(r - r_0)\right]\,\delta\left(\cos\theta - \frac{z_0}{r}\right) \\
\frac{Q}{A} &= [2\pi]\int_{z_0}^{r_0} dr'\, r'^2 \int_1^{\cos\theta_0} d\left(\cos\theta'\right)\,\delta\left(\cos\theta' - \frac{z_0}{r'}\right) \\
\frac{Q}{2\pi A} &= \int_{z_0}^{r_0} dr'\, r'^2 = \frac{1}{3}\left(r_0^3 - z_0^3\right) \\
A &= \frac{3}{2\pi}\,\frac{Q}{r_0^3 - z_0^3}
\end{aligned}$$

leading to

$$\boxed{\rho(r,\theta) = \rho_d\,\left[\Theta(r - z_0) - \Theta(r - r_0)\right]\,\delta\left(\cos\theta - \frac{z_0}{r}\right), \quad \rho_d = \frac{3}{2\pi}\,\frac{Q}{r_0^3 - z_0^3}}$$

The dimensions check out (Q/L^3 for ρ_d) again and we can also confirm that this form has the correct limit

$$\begin{aligned}
&z_0 \to 0, \;\; r_0 \to R, \;\; \theta_0 \to \frac{\pi}{2}, \;\; \rho_d \to \frac{3}{2\pi}\,\frac{Q}{R^3} \\
&\delta\left(\cos\theta - \frac{z_0}{r}\right) = \frac{\delta\left(\theta - \cos^{-1}\frac{z_0}{r}\right)}{\left|\sin\frac{z_0}{r}\right|} \to \delta\left(\theta - \frac{\pi}{2}\right)
\end{aligned}$$

Problem 1.5

The time-averaged potential of a hydrogen atom is given by

$$\Phi = \frac{q}{4\pi\epsilon_0}\frac{e^{-\alpha r}}{r}\left(1+\frac{\alpha r}{2}\right)$$

where $q = |e|$ and $1/\alpha$ is half the Bohr radius a_0. Find the charge distribution (continuous and discrete) that will produce this potential and the physical interpretation.

The charge density $\rho(r)$ can be found using the Poisson equation

$$\rho = -\epsilon_0\,\nabla^2\Phi$$

Since the potential Φ has no dependence on θ or ϕ, we only need the radial part of the Laplacian ∇^2

$$\nabla^2\Phi(r) = \frac{1}{r^2}\partial_r\left[r^2\,\partial_r\Phi(r)\right]$$

but we need to be careful at the origin where the potential blows up and use

$$\nabla^2\left(\frac{1}{r}\right) = -4\pi\,\delta(r)$$

Using the chain rule for the Laplacian

$$\nabla^2\left(\frac{e^{-\alpha r}}{r}\right) = \frac{1}{r}\nabla^2\left(e^{-\alpha r}\right) + 2\vec{\nabla}\left(e^{-\alpha r}\right)\cdot\vec{\nabla}\left(\frac{1}{r}\right) + e^{-\alpha r}\nabla^2\left(\frac{1}{r}\right)$$

and working out the individual terms, we get

$$\begin{aligned}\nabla^2\left(e^{-\alpha r}\right) &= \frac{1}{r^2}\partial_r\left(r^2\,\partial_r e^{-\alpha r}\right)\\ &= -\frac{\alpha}{r^2}\left(2re^{-\alpha r} - r^2\alpha e^{-\alpha r}\right) = \alpha e^{-\alpha r}\left(\alpha - \frac{2}{r}\right)\end{aligned}$$

for the first,

$$\begin{aligned}\vec{\nabla}\left(e^{-\alpha r}\right)\cdot\vec{\nabla}\left(\frac{1}{r}\right) &= \left[\partial_r\left(e^{-\alpha r}\right)\right]\left[\partial_r\left(\frac{1}{r}\right)\right] = \left[-\alpha e^{-\alpha r}\right]\left[-\frac{1}{r^2}\right]\\ 2\vec{\nabla}\left(e^{-\alpha r}\right)\cdot\vec{\nabla}\left(\frac{1}{r}\right) &= 2\alpha\frac{e^{-\alpha r}}{r^2}\end{aligned}$$

for the second, and

$$e^{-\alpha r}\nabla^2\left(\frac{1}{r}\right) = e^{-\alpha r}\left[-4\pi\,\delta(r)\right] = -4\pi\,\delta(r)$$

for the third. Putting these together, we have

$$\begin{aligned}\nabla^2\left(\frac{e^{-\alpha r}}{r}\right) &= \alpha\frac{e^{-\alpha r}}{r}\left(\alpha - \frac{2}{r}\right) + 2\alpha\frac{e^{-\alpha r}}{r^2} - 4\pi\delta(r)\\ &= \alpha^2\frac{e^{-\alpha r}}{r} - 4\pi\delta(r)\end{aligned}$$

The potential has another factor

$$\frac{4\pi\epsilon_0}{q}\Phi = \frac{e^{-\alpha r}}{r}\left(1+\frac{\alpha r}{2}\right) = \frac{e^{-\alpha r}}{r} + \frac{\alpha}{2}e^{-\alpha r}$$

for which we already have the Laplacian

$$\nabla^2 \left(\frac{\alpha}{2} e^{-\alpha r} \right) = \frac{\alpha}{2} \alpha e^{-\alpha r} \left(\alpha - \frac{2}{r} \right) = \alpha^2 e^{-\alpha r} \left(\frac{\alpha}{2} - \frac{1}{r} \right)$$

leading to

$$\begin{aligned} \frac{4\pi\epsilon_0}{q} \nabla^2 \Phi &= \alpha^2 \frac{e^{-\alpha r}}{r} - 4\pi\delta(r) + \frac{\alpha^3}{2} e^{-\alpha r} - \alpha^2 \frac{e^{-\alpha r}}{r} \\ &= -4\pi\, \delta(r) + \frac{\alpha^3}{2} e^{-\alpha r} \end{aligned}$$

and we have the final expression for the charge density

$$\boxed{\rho(r) = -\epsilon_0 \nabla^2 \Phi = q \left[\delta(r) - \frac{\alpha^3}{8\pi} e^{-\alpha r} \right] = \rho_1 + \rho_2}$$

The first term, ρ_1, is a point charge $+|e|$ at the origin representing the nucleus (the proton). The second term represents the electron "cloud". It has total charge

$$\int d^3x \, \rho_2(x) = 4\pi \int_0^\infty dr \, r^2 \rho_2(r) = -\frac{q\alpha^3}{2} \int_0^\infty dr \, r^2 e^{-\alpha r}$$

We will keep encountering this integral and a simple trick to evaluate it is differentiation under the integral sign

$$\int_0^\infty dr \, r^2 e^{-\alpha r} = \partial_\alpha^2 \left[\int_0^\infty dr \, e^{-\alpha r} \right] = \partial_\alpha^2 \left(\frac{1}{\alpha} \right) = \frac{2}{\alpha^3}$$

Thus, the total charge associated with the 2nd term is

$$\int d^3x \, \rho_2(r) = -q = -|e|$$

as expected for a single electron. Note that this charge density is related to the ground state wave function for the electron from the Schrödinger equation through

$$\rho_2(r) = q \, |\psi_{1s}(r)|^2$$

since $|\psi(r)|^2$ is the probability density for the electron with the wavefunction

$$\psi_{1s}(r) = \frac{e^{-r/a_0}}{\sqrt{\pi a_0^3}}$$